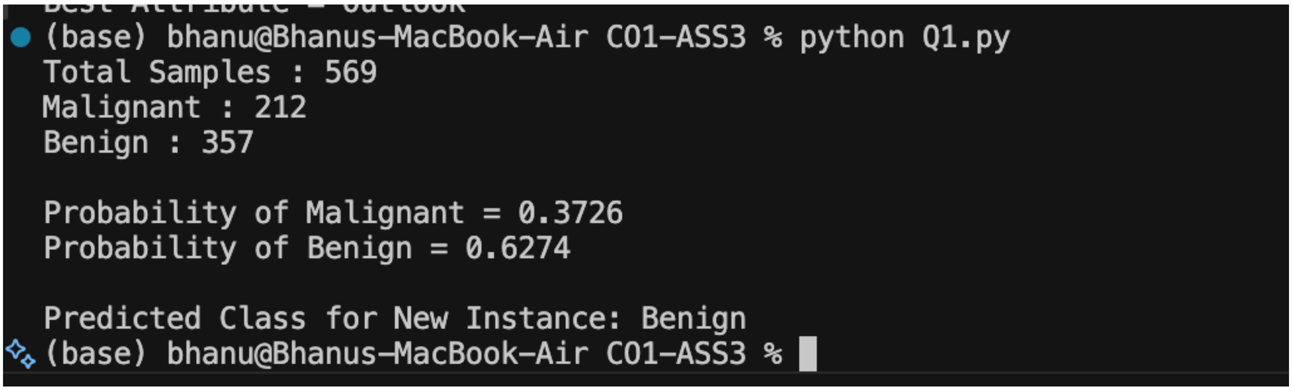
Experiment 1: Probability Theory



```
(base) bhanu@Bhanus-MacBook-Air C01-ASS3 % python Q1.py
Total Samples : 569
Malignant : 212
Benign : 357

Probability of Malignant = 0.3726
Probability of Benign = 0.6274

Predicted Class for New Instance: Benign
(base) bhanu@Bhanus-MacBook-Air C01-ASS3 %
```

Result

The Breast Cancer Wisconsin Diagnostic Dataset was successfully loaded and analyzed. The number of malignant and benign cases was counted, and the prior probability of each class was calculated.

The computed probabilities were:

- Probability of Malignant = **0.3726**
- Probability of Benign = **0.6274**

Since the probability of the Benign class is higher than the Malignant class, the new data instance was predicted as **Benign** based on the prior probabilities.